Inflammation & Cancer

Melissa R. Weege, MS, RTT, CMD
University of Wisconsin- La Crosse

Purpose & Objectives

- The purpose of this session will be to educate radiation therapists about the role of inflammation and its propensity to cause neoplastic disease. Additionally, participants will be educated on the importance of preventing and treating inflammatory conditions in order to reduce the likelihood of cancer incidence.

Objectives:

- Following the session, participants will be able to:
- Identify causes of inflammation in the body.
- Describe how inflammation manifests itself the body.
- Understand the role that inflammation plays in cancer causation.
- Describe how the anti-inflammatory diet and other factors help to reduce inflammation.
- Identify the importance of eliminating inflammation as a healthy radiation therapist.

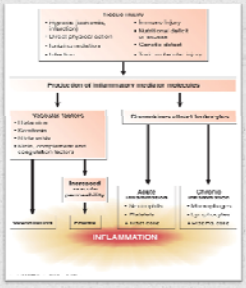
What is inflammation?[1]

- Non specific response to injury
- Will occur anytime there is an interruption of tissue homeostasis [2]
- Protective mechanism
- Beginning of the healing process
- Includes a cellular and vascular response

Causes of Inflammation[1]

- Infection
- Irritations
- Immune injury
- Direct physical trauma or injury
- Genetic defects
- Ionizing radiation
- Nutritional deficits or excess
- Toxic chemicals
- Hypoxia
- Obesity

Overview of Inflammation[1]

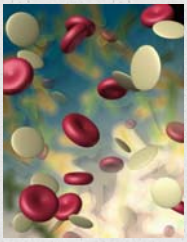


Vascular Response[1]

1. Injury or insult to tissue (chemical mediators released)
2. Arterioles vasoconstrict to ↓ blood loss
3. Precapillary arterioles vasodilate to ↑ blood flow to damaged tissue
4. Increased permeability
5. Leakage of fluid to extravascular space
6. Edema

Cellular Response[1]

- Phagocytosis
  - Polys
  - Monocytes
  - Macrophages
- Inflammatory cells
  - Polys
  - Monocytes
  - Eosinophils
  - Macrophages



Acute vs. chronic inflammation[1]

- Result of short term injury
- Lasts few hours or days
- Characterized by:
  - Accumulation of fluid
  - Infiltration of neutrophils
  - Vascular dilation
  - Resolution without scarring

- Result of long term or persistent injury
- Last weeks to years
- Characterized by:
  - Fibrosis
  - Lymphocytes, macrophages, and plasma cells
  - Ingrowth new blood vessels

Review of Carcinogenesis[1]

- Carcinogens
  - Viruses, infections, toxins, chronic inflammation
- Initiation
  - Carcinogen causes a DNA mutation
- Promotion
  - Carcinogen that proliferates a mutated cell
- Angiogenesis
  - Tumor develops its own blood system

Chronic Inflammation [2,3]

- Free radicals
- Cell mediators, chemokines, cytokines, natural killer cells

Inflammation & Cancer [2,3,4]

- Reported by Virchow in early 1863
- Long term chronic inflammation leads to dysplasia
  - Hepatitis B & C
  - HPV
  - H. pylori
  - Inflammatory bowel disease
  - Chronic Pancreatitis
  - Several caused by inflammation alone

Inflammation based tumors[3]

- Not an infection or irritation causation
- Pathology shows:
  - Lymphocytes
  - Fibroblasts
  - Macrophages
- Lower or less optimistic treatment outcomes and prognosis

Examples of Inflammation-based cancers[3]

- Hepatocellular
- Cervical cancer
- Colon cancer
- Stomach cancer
- Pancreatic cancer
- Esophageal

Obesity & Inflammation[1,4]

- Obesity defined- Body Mass Index
  - Based on height and weight, & gender
  - Weight in pounds * 704.5/ height in inches^2
  - BMI over 30 considered obese
  - According to NHI: http://win.niddk.nih.gov/statistics/index.htm
- Adipose tissue creates low grade, whole body chronic inflammation.
- Risk: Cardiovascular disease, diabetes, and obesity related cancers

Obesity as a carcinogen[4]

- How obesity can cause cancer:
  - Fat produces extra estrogen
  - Increased levels of insulin
  - Fat creates additional hormones which will stimulate cell growth
  - Chronic low level inflammation

Obesity caused caners[4]

- Breast
- Endometrial
- Colorectal
- Kidney
- Esophageal
- Pancreatic
- Thyroid
- Gallbladder
- Potential link with Prostate

What signs & symptoms indicate occurrence of inflammatory processes [1, 4, 5]

- Runny nose, scratchy throat
- Skin rashes
- Joint pain
- Headache
- Fatigue
- Digestive issues
- Cholesterol
- Autoimmune disease
- Elevated C-RP levels

http://www.medicalnewstoday.com/articles/248423.php

How can we minimize chronic inflammation?[4,5]

- Weight management
- Nutrition
  - Anti-inflammatory foods
  - Supplementation

Weight Management

- Dietary Modification- Calorie Reduction
  - Small changes make big gains in health improvement [5]
- Exercise
  - People who are fit have lower levels of inflammatory markers[5]

Nutrition

- Anti-inflammatory diet
- Cancer Fighting foods
- Supplementation

What is the #1 Dietary Cause of Cancer?[7]

Sugar

Foods known to cause inflammation

- Alcohol
- Saturated Fats
- Gluten
- Dairy
- Diet Soda
- MSG

Foods known to cause cancer[7]

- Processed meats
- Acrylamides
- Artificial sweeteners
- Soy

Anti-inflammatory Diet

- Disease prevention & reduction of inflammatory symptoms
- Foods high in Omega 3 fats, antioxidants, high in fiber, spices with anti-inflammatory compounds
- Avoid foods high in trans and omega-6 fats, high in simple carbohydrates, those known to trigger intolerance.

Cancer Prevention Foods[7]

- Green Tea
- Broccoli
- Cabbage & Cauliflower
- Dark, Leafy Greens
- Fermented Soy
- Citrus Fruits
- Berries
- Nuts
- Avocados
- Tomatoes

Summary of nutrition[7]

- Increase healthy fats, decrease bad fats
- Adequate amounts of protein at each meal
  - Men: 20 grams
  - Women: 15 grams
  - Post workout: 5-10 grams
- Eat 8-10 servings fruits and veggies
- Eliminate sugars
- Take basic supplements

What a meal should look like…

- http://www.choosemyplate.gov

Supplementation

- Supplementation
  - Fill nutritional deficits
    - RDA- Recommended Daily Allowance
- Supplement quality

Multivitamin

- Provides core nutrition of vitamins and minerals
- Fills in missing gaps from diet
- Best when consumed with food

Omega 3's[8]

- DHA & EPA
- Naturally reduces inflammation and anxiety
- Found in fish (sardines, herring, mackerel)
- 1000mg per day to reduce inflammation, weight management, prevent heart disease & cancer.
  - Dr. Carl Lavie reports cardioprotection, small waist and hip ratios, and lower weight

http://supplementalscience.wordpress.com/2009/09/30/good-news-just-keeps-mounting-for-omega-3s/

Probiotics

- Gut flora- microbiota[9]
- What are they?
- Benefits:
  - Digestion
  - Immunity, protection from stress & environment
  - Female Health
  - Weight Management, nutrient absorption
- Reduce cancer risk by stimulating immune system and deactivating cancer causing enzymes[7]

Probiotics

- Optimal is choosing a good supplement and consuming them on an empty stomach
- 6 billion cfu and 6 different strains- high quality
- Lactobacillus rhamnosus, Lactobacillus acidophilus, Lactococcus lactis ssp. Lactis, Bifidobacterium bifidum, Bifidobacterium longum, Lactobacillus delbrueckii ssp. Bulgaricus and Streptococcus thermophilus, fructooligosaccharides.

Case Study

	Normal	April, 2009	September 2009	September 2012
Total Cholesterol	< 200	201	188	164
LDL	<130	140	130	104
HDL	40-60	32	35	39
Triglycerides	38-150	147	116	102
Glucose	70-99	101	96	87

Why does this all matter?

- Costs of obesity, heart disease, and cancer
- How health care workers are often unhealthy
- Small changes over the course of time so that you stick with it. You will be amazed at how much you change over time.
- Not same advice for patients undergoing treatment so make sure they see a dietitian.

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